

VisionForge: A Modular Classical Computer Vision Architecture for Multistage Image Processing, Spatial Feature Extraction, and Video Scene Dynamics

Design, Mathematical Formulations, and Empirical Latency Evaluation

Parth Jangir

Registration No.: 24BAI10042 | Department of Computer Science & Engineering
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Abstract

Automated visual inspection and scene interpretation require deterministic, low-latency, and interpretable algorithm pipelines that operate reliably without black-box neural dependencies or high-end GPU hardware. This report presents **VisionForge**, an end-to-end Python framework that unifies fundamental classical computer vision routines into a cohesive five-stage architecture: (1) low-level image preprocessing and frequency-domain spectral transformation; (2) multi-scale feature extraction encompassing edge, corner, scale-space, and Gabor texture descriptors; (3) spatial and color-space image segmentation; (4) geometric shape detection, contour analytics, and pattern classification; and (5) video-based scene dynamics incorporating background subtraction, dense/sparse optical flow, and affine motion estimation.

We provide complete mathematical derivations for every pipeline component—from Contrast-Limited Adaptive Histogram Equalization (CLAHE) and 2D Discrete Fourier Transforms to Harris second-moment matrices, K-Means color clustering, Hough space accumulators, and Lucas-Kanade gradient constraints. In benchmark trials on $512 \times 512 \times 3$ image arrays, VisionForge executed the full static analysis sequence in 8.3015 s, extracting 364 Harris corner keypoints, 142,884 HOG descriptors, 23 Hough lines, 31 circles, and 28 closed contours without triggering unhandled runtime exceptions. Experimental evaluation confirms that deterministic feature engineering delivers transparent, parameter-tunable visual representations suitable for real-time edge processing and autonomous system telemetry.

1 Introduction and Theoretical Motivation

Computer vision applications across industrial automation, robotics, and remote sensing increasingly depend on deep learning architectures. While deep convolutional networks and vision transformers excel at high-level semantic classification, they exhibit notable drawbacks in constrained engineering environments: substantial memory footprints, black-box decision boundaries, high computational overhead, and susceptibility to adversarial perturbations. Classical computer vision techniques, grounded in signal processing, differential geometry, and statistical estimation, provide deterministic execution guarantees and direct interpretability.

The primary objective of **VisionForge** is to synthesize classical vision methodologies into a unified, modular execution pipeline. Rather than executing isolated algorithms, VisionForge chains low-level pixel transformations into higher-order geometric and dynamic representations. The framework addresses key operational constraints:

- **Hardware Independence:** Pure CPU execution utilizing OpenCV, NumPy, SciPy, scikit-learn, and scikit-image without CUDA or GPU tensor dependencies.
- **Pipeline Determinism:** Traceable mathematical transitions between raw BGR arrays, frequency spectra, edge probability maps, segmented regions, and kinematic motion vectors.
- **Comprehensive Telemetry:** Standardized JSON metadata logging and automated performance evaluation tracking per-stage latency and spatial feature statistics.

2 System Architecture and Implementation

The VisionForge framework is structured into five decoupled Python modules within the `src/` directory, controlled by a centralized pipeline orchestrator (`src/pipeline.py`) and accessible via a unified command-line interface (`main.py`).

2.1 Module 1: Preprocessing Engine and Spectral Analysis

Raw sensor images often suffer from non-uniform illumination, sensor thermal noise, and low local contrast. Module 1 (`src/preprocessing.py`) conditions input arrays before spatial derivative evaluation.

2.1.1 Luminance Reduction and Noise Attenuation

Input BGR color images $I_{\text{BGR}} \in \mathbb{R}^{H \times W \times 3}$ are converted to single-channel grayscale $Y \in \mathbb{R}^{H \times W}$ following the ITU-R BT.601 standard:

$$Y(x, y) = 0.299 \cdot R(x, y) + 0.587 \cdot G(x, y) + 0.114 \cdot B(x, y) \quad (1)$$

To attenuate high-frequency additive noise while preserving macro structure, we apply a 2D isotropic Gaussian filter parameterized by kernel size $k \times k$ and standard deviation σ :

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (2)$$

For impulsive (salt-and-pepper) noise, a median filter replaces pixel $Y(x, y)$ with the median of its $k \times k$ neighborhood $\mathcal{N}(x, y)$, preserving sharp edge boundaries.

2.1.2 Adaptive Contrast Enhancement (CLAHE)

Global histogram equalization frequently over-amplifies noise in homogeneous regions. VisionForge implements Contrast-Limited Adaptive Histogram Equalization (CLAHE). The image is partitioned into non-overlapping contextual tiles (8×8). Within each tile, the local intensity histogram is clipped at threshold $\beta = 2.0$, redistributing excess pixels uniformly across all bins prior to cumulative distribution function (CDF) calculation:

$$y_{\text{new}} = \left(\frac{\text{CDF}(y) - \text{CDF}_{\min}}{(W_{\text{tile}} \cdot H_{\text{tile}}) - \text{CDF}_{\min}} \right) \times (L - 1) \quad (3)$$

Bilinear interpolation across adjacent tile boundaries eliminates artificial block artifacts.

2.1.3 Frequency Domain Fourier Spectrum

To analyze structural periodicities and directional textures, we compute the 2D Discrete Fourier Transform (DFT):

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \exp\left(-j2\pi \left(\frac{ux}{M} + \frac{vy}{N}\right)\right) \quad (4)$$

The origin is shifted to the spectrum center, yielding the normalized logarithmic magnitude spectrum $M(u, v) = \log(1 + |F(u, v)|)$.

2.2 Module 2: Feature Extraction and Scale-Space Engine

Module 2 (`src/feature_extraction.py`) computes local invariants, multi-scale scale-space representations, and high-dimensional texture descriptors.

2.2.1 Multi-Scale Edge Detection

We incorporate three edge detection formulations:

1. **Canny Edge Detector:** Calculates gradient magnitude $|\nabla Y| = \sqrt{I_x^2 + I_y^2}$ and direction $\theta = \arctan(I_y/I_x)$. Non-maximum suppression thins candidate edges, followed by dual-threshold hysteresis. Thresholds are dynamically assigned using median intensity heuristics:

$$T_{\text{low}} = \max(0, 0.66 \cdot \text{med}(Y)), \quad T_{\text{high}} = \min(255, 1.33 \cdot \text{med}(Y)) \quad (5)$$

2. **Laplacian of Gaussian (LoG):** Identifies zero-crossings in second-order derivatives:

$$\nabla^2 G(x, y; \sigma) = \frac{x^2 + y^2 - 2\sigma^2}{2\pi\sigma^6} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (6)$$

Responses are scale-normalized by σ^2 to enable cross-scale blob comparison.

3. **Difference of Gaussians (DoG):** Approximates LoG via bandpass subtraction of Gaussian blurred images at scales σ_1 and $\sigma_2 = k\sigma_1$:

$$\text{DoG}(x, y) = G(x, y; \sigma_1) * Y(x, y) - G(x, y; \sigma_2) * Y(x, y) \quad (7)$$

2.2.2 Harris Corner Detection

Salient interest points are located by constructing the local auto-correlation second-moment matrix $\mathbf{M}$:

$$\mathbf{M}(x, y) = \sum_w \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \quad (8)$$

where I_x, I_y are spatial Sobel derivatives. The corner response metric R avoids explicit eigenvalue computation:

$$R = \det(\mathbf{M}) - k \cdot (\text{trace}(\mathbf{M}))^2 = (\lambda_1 \lambda_2) - k(\lambda_1 + \lambda_2)^2 \quad (9)$$

with empirical constant $k = 0.04$. Non-maximum suppression isolates localized corner peaks.

2.2.3 Gabor Filter Bank and HOG Descriptors

Texture analysis utilizes a Gabor filter bank spanning orientation angles $\theta \in \{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\}$:

$$g(x, y; \lambda, \theta, \psi, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cos\left(2\pi \frac{x'}{\lambda} + \psi\right) \quad (10)$$

where $x' = x \cos \theta + y \sin \theta$ and $y' = -x \sin \theta + y \cos \theta$.

For object representation, the Histogram of Oriented Gradients (HOG) pipeline divides images into 8×8 pixel cells, accumulates gradient directions across 9 angular bins, and normalizes feature vectors over 2×2 cell blocks using the L2-norm:

$$\mathbf{v}_{\text{norm}} = \frac{\mathbf{v}}{\sqrt{\|\mathbf{v}\|_2^2 + \epsilon^2}} \quad (11)$$

2.3 Module 3: Image and Spatial Segmentation

Module 3 (`src/segmentation.py`) partitions input images into distinct homogeneous semantic regions.

2.3.1 Color-Space K-Means Clustering

Pixels are mapped into 3D color feature space (Lab or BGR) and partitioned into K clusters by minimizing the total within-cluster sum of squares (WCSS):

$$J = \sum_{j=1}^K \sum_{\mathbf{x}_i \in S_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2 \quad (12)$$

where $\boldsymbol{\mu}_j$ denotes the centroid of cluster S_j . Centroids iterate until convergence $\|\boldsymbol{\mu}_j^{(t+1)} - \boldsymbol{\mu}_j^{(t)}\| < \delta$.

2.3.2 Mean-Shift and Watershed Segmentation

- **Mean-Shift:** Performs non-parametric density gradient estimation in joint spatial-color space (x, y, L, a, b) , shifting window seeds towards local modes.
- **Region Growing:** Iteratively aggregates adjacent pixels starting from seed points based on intensity difference criteria $|Y(x, y) - Y_{\text{seed}}| \leq \tau_{\text{region}}$.
- **Marker-Based Watershed:** Treats inverted gradient magnitude images as topographic surfaces. Distance transform local maxima establish initial catchment basin markers, preventing over-segmentation.

2.4 Module 4: Shape Detection and Pattern Recognition

Module 4 (`src/pattern_analysis.py`) converts segmented regional masks into parametric geometric primitives and feature vectors.

2.4.1 Hough Parametric Transforms

- **Hough Line Transform:** Maps edge points (x, y) into polar accumulator space (ρ, θ) :

$$\rho = x \cos \theta + y \sin \theta \quad (13)$$

Accumulator cell peaks exceeding threshold T_{hough} identify collinear edge segments.

- **Hough Circle Transform:** Employs a 3D accumulator array (a, b, r) corresponding to the parametric circle equation $(x - a)^2 + (y - b)^2 = r^2$. Gradient directions guide accumulator voting to reduce search complexity.

2.4.2 Contour Properties and Dimensionality Reduction

For closed contours identified via Suzuki's algorithm, geometric invariants are extracted:

$$\text{Circularity } \mathcal{C} = \frac{4\pi \cdot \text{Area}}{\text{Perimeter}^2}, \quad \text{Solidity } \mathcal{S} = \frac{\text{Area}}{\text{Convex Hull Area}} \quad (14)$$

High-dimensional shape descriptor vectors are projected onto lower-dimensional orthogonal axes via Principal Component Analysis (PCA) by solving the covariance matrix sample decomposition $\Sigma \mathbf{v} = \lambda \mathbf{v}$. Pattern classification is performed using K-Nearest Neighbors (KNN) under Euclidean distance metrics.

2.5 Module 5: Video Motion Engine and Scene Dynamics

Module 5 (`src/motion_analysis.py`) processes consecutive video frames to quantify motion field dynamics.

2.5.1 Background Subtraction

Dynamic scene background models employ Gaussian Mixture Models (MOG2). The probability of observing pixel intensity $\mathbf{x}_t$ at time t is modeled as:

$$P(\mathbf{x}_t) = \sum_{i=1}^K w_{i,t} \cdot \eta(\mathbf{x}_t; \boldsymbol{\mu}_{i,t}, \boldsymbol{\Sigma}_{i,t}) \quad (15)$$

where $w_{i,t}$ are mixture weights and η represents the Gaussian density. Parameters update recursively using adaptive learning rate α .

2.5.2 Dense and Sparse Optical Flow

- **Dense Optical Flow (Farnebäck)**: Approximates local neighborhood signal intensities with quadratic polynomials $f(\mathbf{x}) \sim \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$, tracking polynomial displacement under spatial transformations.
- **Sparse Optical Flow (Lucas-Kanade / KLT)**: Assumes constant velocity within localized windows Ω , solving the brightness constancy spatial gradient constraint:

$$\begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix} \begin{bmatrix} V_x \\ V_y \end{bmatrix} = - \begin{bmatrix} \sum I_x I_t \\ \sum I_y I_t \end{bmatrix} \quad (16)$$

Global motion vectors fit a 6-parameter 2D affine transformation matrix yielding frame translation (t_x, t_y) , rotation angle θ , and scale factor s .

3 Experimental Benchmark and Evaluation

We evaluated VisionForge using a standardized $512 \times 512 \times 3$ pixel benchmark image (`data/images/sample.jpg`). Processing timings and feature counts were measured under Python 3.9 execution on an Intel Core i7 environment.

Table 1: VisionForge Pipeline Stage Processing Latency Profile (512×512 Input)

Pipeline Stage	Sub-Operation / Algorithm	Execution Time (s)	% Total
Module 1: Preprocessing	Grayscale Conversion	0.0020	
	Gaussian Blur	0.0010	
	Histogram Equalization	0.0000	
	Contrast Enhancement (CLAHE)	0.0890	
	Histogram Analysis	0.0903	
	2D Fourier Spectrum (FFT)	0.4107	
Module 2: Feature Extraction	Canny Edge Detection	0.0020	
	Laplacian of Gaussian (LoG)	0.0091	
	Difference of Gaussians (DoG)	0.0039	
	Harris Corner Detection	0.0041	
	HOG Descriptor Generation	0.2715	
	Scale-Space Representation	0.4078	
	Gabor Filter Bank	0.6864	
Module 3: Segmentation	K-Means Color Clustering ($K = 5$)	2.0897	
Module 4: Shape Analysis	Hough Line Transform	0.0172	
	Hough Circle Transform	0.0160	
	Contour Geometry Analysis	0.0040	
Total Static Pipeline Execution	All 5 Modules Combined	8.3015	

Table 2: Quantitative Spatial Feature Extraction Summary

Metric Category	Parameter / Descriptor	Measured Value
Image Statistics	Mean Intensity / Std Deviation	82.37 / 61.81
	Dominant Frequency (FFT Spectrum)	9.00 cycles/px
Boundary Analysis	Canny Edge Density Ratio	0.0300 (3.00%)
Keypoint Detection	Harris Corner Count ($k = 0.04$)	364 corners
HOG Descriptor	Feature Vector Dimension	142,884 elements
Geometric Primitives	Detected Hough Lines	23 lines
	Detected Hough Circles	31 circles
	Closed Contours Extracted	28 contours

4 Practical Engineering Observations and Limitations

During implementation and experimental testing, several practical trade-offs emerged:

1. **K-Means Clustering Computational Bottleneck:** As documented in Table 1, K-Means color clustering ($K = 5$) accounted for 25.17% (2.0897 s) of total execution latency. While iteration convergence is deterministic, processing high-resolution images benefit significantly from spatial downsampling prior to color-space distance calculations.
2. **Gabor Filter Bank Overhead:** Convolution image arrays with multi-orientation Gabor filter banks consumed 0.6864 s. In resource-constrained edge deployments, restricting Gabor evaluation to pre-selected edge ROI masks reduces redundant spatial convolutions.
3. **Sensitivity to Dynamic Lighting:** Classical photometric thresholding and Watershed marker initialization exhibit sensitivity to severe non-uniform shadows. Incorporating

CLAHE in Module 1 stabilizes downstream Canny hysteresis boundaries under varying sunlit conditions.

5 Conclusion

VisionForge demonstrates a robust, complete classical computer vision architecture capable of performing multi-stage image processing, multi-scale feature extraction, spatial segmentation, shape analysis, and video motion tracking. By avoiding deep neural networks, the system operates deterministically on CPU hardware, executing complete static pipeline evaluations in 8.3015 s while logging comprehensive mathematical telemetry. A comprehensive unit test suite (`pytest tests/ -v`) containing over 20 synthetic test cases verifies zero runtime exception rates, confirming VisionForge's suitability for reliable visual inspection tasks.

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